

Design of Inverter-Based Ring Oscillators

Chapter 3 Razavi PLL

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Presentation Topics

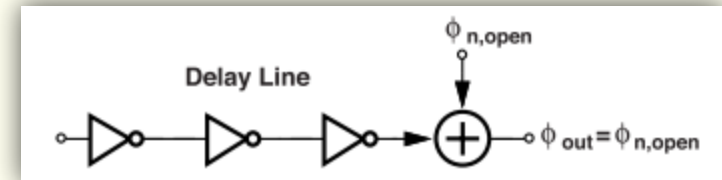
1. Phase Noise in Ring Oscillators
2. Preliminary Design Ideas
3. Obtaining the Desired Oscillation Frequency – 4 techniques
4. Four 2-GHz oscillator designs.
5. Frequency tuning
6. Continuous and discrete tuning
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9. Discrete tuning
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Phase noise in Ring oscillators

Establish a relationship b/w Open & Closed loop phase noise of a Ring Osc

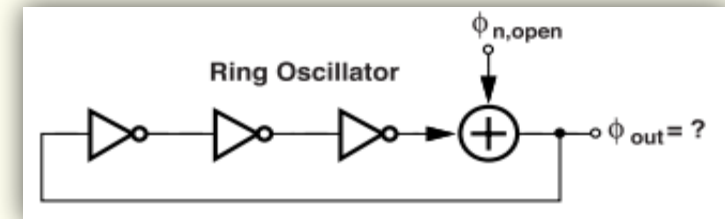
Open loop phase noise:

- Each stage in the Delay Line corrupts the signal only once.
- $\Phi_n = \Phi_{out}$; no loop



Closed loop Phase noise:

- Noise circulates in the loop and the signal is corrupted multiple times.
- This accumulates jitter indefinitely.
- Φ_n is not equal to Φ_{out}



Open Loop and closed Loop Phase noise equations

- **Ring oscillator with a total delay of TDL**

- Delay of one inverter = TD
- Total oscillation delay = $T_{osc} = 6TD$
- TDL has a phase shift of :

$$2\pi (TDL / T_{osc}) = 180 \text{ degrees}$$

- $TDL = T_{osc}/2 = 1/(2f_0)$;

- Add a constant phase of ϕ_1 :

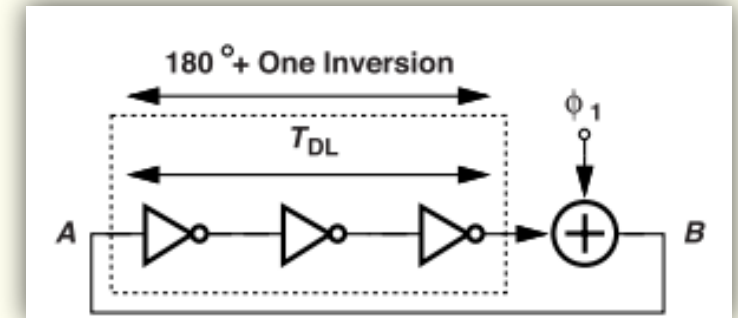
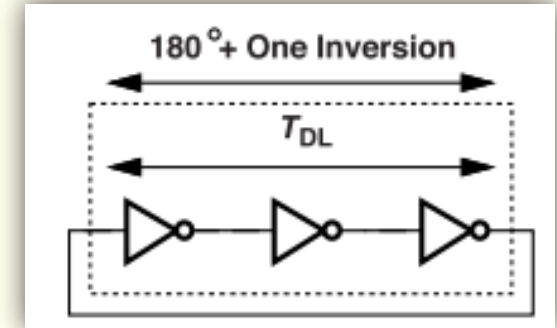
- Phase shift has to be 180 degrees for oscillations

- **Frequency Adjustment :**

- $2\pi (TDL / T_{osc}) + \phi_1 = \pi$
- **New freq of oscillation is : $f_{osc} = f_0 - f_0 * \{\phi_1 / \pi\}$**
- Freq of oscillation without added phase = f_0

Osc Freq is dependent of the phase shift.

Gate delays TD do not change.

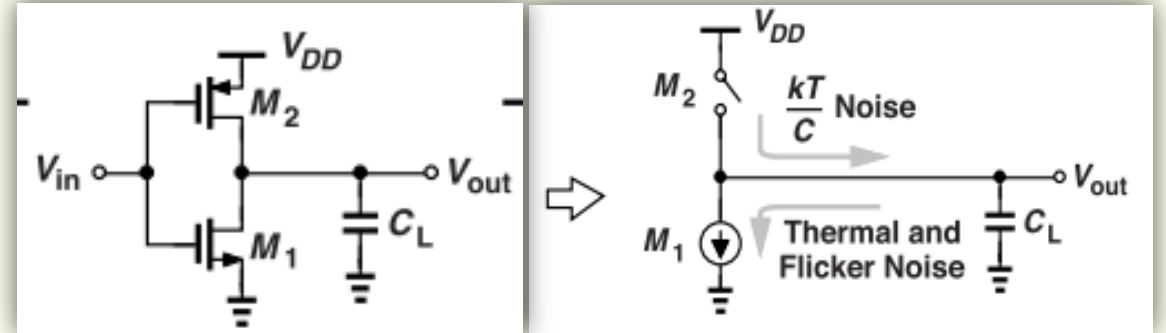


Noise sources in CMOS inverter

1. NMOS and PMOS produce : Thermal (white) noise and Flicker noise.
2. Falling edge in V_{out} is corrupted.
3. Noise from NMOS and PMOS is injected into the capacitance C_L .

WORKING

- NMOS (M_1) injects:
 - Thermal noise
 - Flicker noise ($1/f$)
- PMOS (M_2) turns off suddenly:
 - Leaves kT/C noise on load capacitor C_L



***These noise sources corrupt the output edge,
causing timing jitter → which translates to phase noise in oscillation.***

Phase Noise in CMOS Inverter Oscillators

Analysis result in N stage ring oscillator:

○ White noise :

$$S_{\phi n, white}(f) = \frac{f_0^2}{f^2} \cdot \frac{4kT\gamma}{I_D(V_{DD} - |V_{TH}|)}$$

○ Flicker noise :

$$S_{\phi n, flicker}(f) = \frac{f_0^2}{f^2} \frac{1}{4NI_D^2} (S_{1/f, NMOS} + S_{1/f, PMOS})$$

Assumptions:

- $g_{m,N} = g_{m,P}$
- $I_{DN} = I_{DP}$
- $V_{thn} = |V_{thp}|$

○ **To be observed:**

▪ **White noise PN :**

- Independent of Number of stages (N) & Capacitance CL
- Improved by higher ID, VDD, and lower VTH

▪ **Flicker Noise Phase Noise :**

- Dependent inversely on CL(Nmos & Pmos) but independent of N.
- Improved by higher ID, more stages N

Phase noise of N-stage Ring Oscillator calculated using Phase noise of 1 inverter

Effects of Variations in the width on PN

If we **double** the widths of all the transistor;

1. Oscillator frequency remains Same
 - Capacitance is increased; Strength is increased ; Delay/stage is Same
2. Phase Noise :
 - White noise is Halved
 - Since, g_m and I_D doubles --> Noise current doubles
 - Flicker noise is Halved
 - Gate area increases

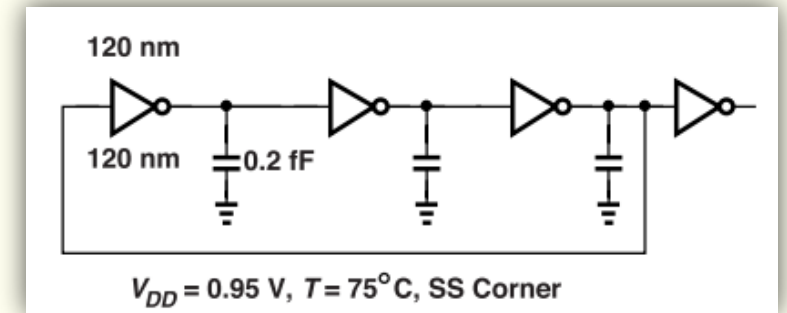
Greater width translates to lower phase noise

Phase noise decreases --> Power consumption increases

Preliminary Design Ideas

1. Initial design :

- 3-stage ring oscillator
- Minimum-size devices {W/L = 120nm/ 40nm} & VDD= 1V
- **Results :**
 - ~45GHz (very high and ideal conditions)
 - No full swings (due to fast feedback).
- Problem with the design:
 - Doesn't include interconnect capacitances,
 - buffer load & Worse PVT case.



2. Worst case Scenario:

- Interconnect CL=0.2 fF ; Buffer (fanout = 1)
- VDD = 0.95V, T = 75°C, SS corner {slow VDD, High Temp}
- Results: Frequency drops to ~21 GHz

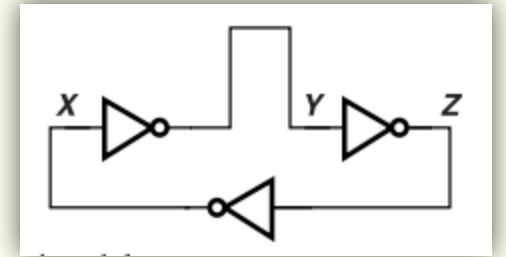
Simulations Done for Worse-case :

Improved Layout

- Reduces wire capacitance between last and first inverter.
- Equalizes delays by symmetrical layout + wires

Strengthened PMOS

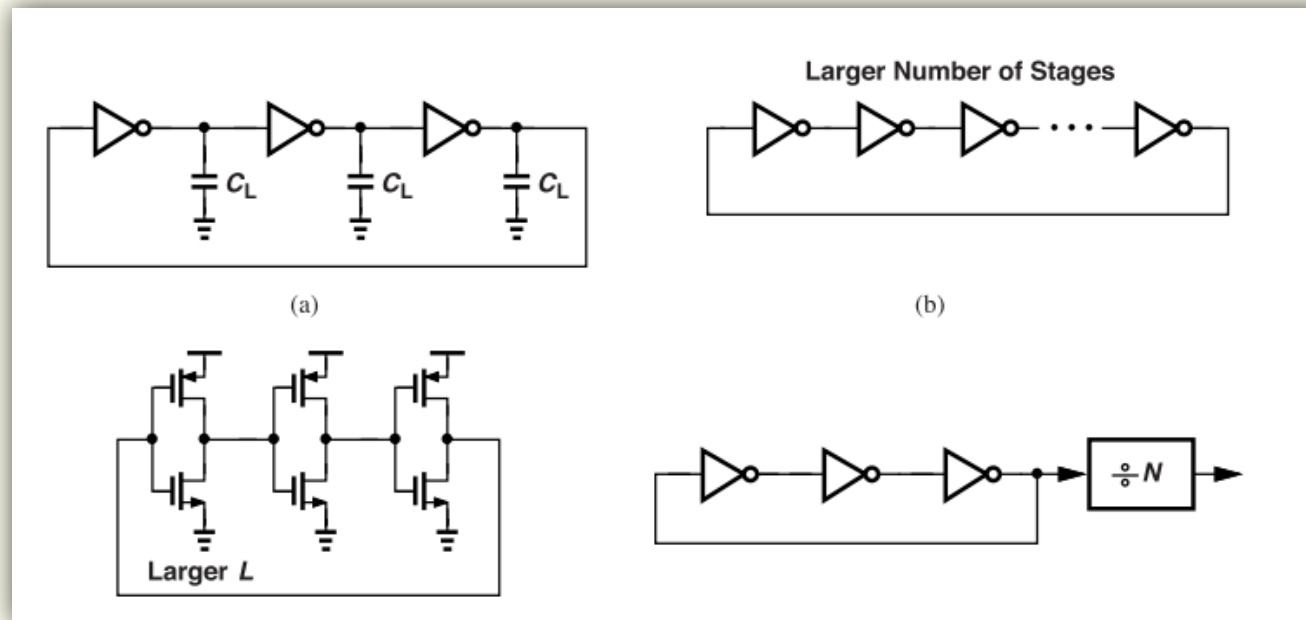
- PMOS is doubled ;
 - Drive current is increased.
 - PMOS **charges the load capacitance CL** faster.
 - Better pull up {the voltage closer to VDD}
-
- Frequency increases to ~22.6 GHz
 - The faster CL charge lowers the delay slightly



Obtaining the Desired Frequency

EX: Slowing Down the Ring Oscillator to 2 GHz

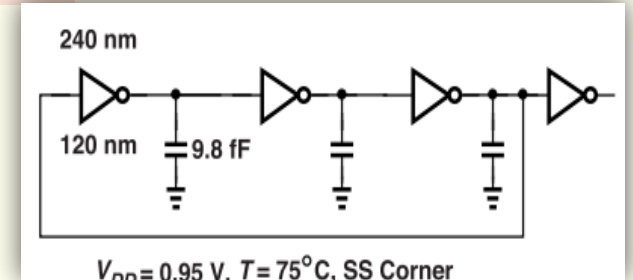
- a) Add capacitance to each node.
- b) Increase the number of stages.
- c) Increase the length of the transistors.
- d) Divide the output frequency by a factor of 10 or 11.



a. Add capacitance to each node.

Reduce oscillation frequency from **22.6 GHz** → **~1.93 GHz**

1. **Capacitance** per node : From 0.93 fF → ~10.5 fF
 - increases gate delay → slows down oscillation.
2. **Power** consumption stays the same : $P = 3 * f_0 * C_{tot} * (V_{DD}^2)$
 - f_0 decreases & C_{tot} increases = Nullify
3. Supply sensitivity K_{vdd} and Normalised value.
 - **K_{Vdd}** decreases; lower absolute freq -> less sensitive in GHz/V terms.
 - **K_{VDD}/f_0** fairly remains the same.



b. Add capacitance to each node.

Increase inverters from **N = 3** to **31** : Oscillations of **2.3Ghz** with **Sharper Edges**

1. Each inverter still sees **unity fanout** → rise/fall times preserved.
2. Power is Independent of the stage count. $P = \{C_{tot} * (V_{dd}^2)\} / 2TD$; $f_0 = 1/2NTD$
3. K_{Vdd} and K_{Vdd}/f_0 remain same.

C. Higher transistor lengths

Increase Length to **L=240nm** to obtain a frequency of **~2.5GHz**

1. Increasing L weakens the Inverters : Less current drive; switching is slow
2. Input **Capacitance** rises due to gate area increase
3. Flicker noise decreases: due to larger channels have less Low-freq noise.
 - **Delay is proportional to L^2**
4. **Power** consumption is reduced: weaker Inverters-> weaker IDD
5. Rounded peaks : due to weaker inverters : less pull.

d. Frequency division

Adding a 4th inverter to the o/p of the ring oscillator

1. This acts as a frequency divider/counter.
2. Counter consumes more power
3. No change in sensitivity or KV_{dd}/f_0 values

Phase Noise Considerations

- We know that Power is inversely proportional to phase noise
- And phase noise directly affects the oscillation frequency of the ring osc.

Example: Reducing Phase noise by 3dB (approx a factor of 2)

- a) **Adding** identical oscillator outputs. Combined phase noise is HALVED.
- b) **Scaling up** width and capacitances; scaling down R and L;
 - Voltage swing & frequency (RC & LC don't change) remains same.
 - Phase noise is HALVED.
 - *You get lower phase noise but use more power.*

Transistor Noise Simulation

- Transistor noise contributes to Phase noise :
 - Thermal noise (is flat across freq range).
 - Flicker noise (dominant at low freq; 1/f fall).
- Flicker noise corner in Ring Osc.
 - We have considered contributions of NMOS devices.
 - Oscillator's flicker noise corner is LOWER than of a single transistor by a factor of $2N$. { $S_{1/f,NMOS} = (1 / 2N) * S_{1/f,NMOS}$ }

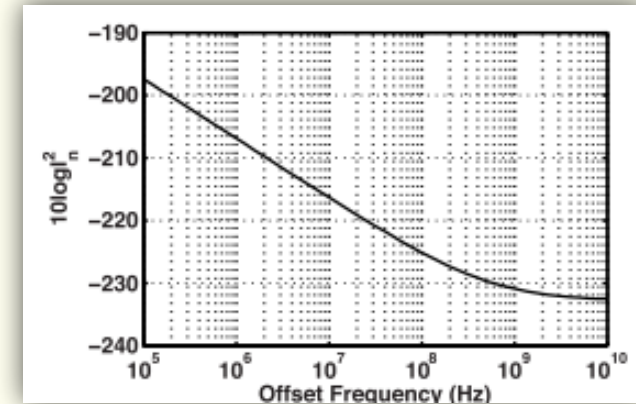
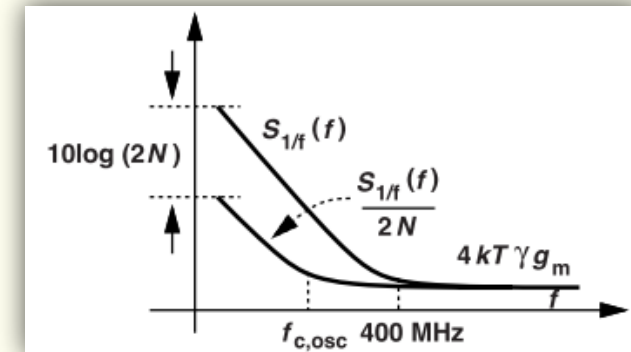


Figure of Merit

- Offset : f at which the phase noise is defined.
- We report $10\log\text{FOM}$
- FOM improves as we go from the 1/f-noise regime (low f) to the white-noise regime (high f) as phase noise reduces.



$$\text{FOM} = \frac{1}{\text{phase noise} \times \text{power dissipation}} \left(\frac{f_0}{f} \right)^2$$

Four 2-GHz oscillator designs.

How do design ideas previously proposed affect phase noise and FOM?

a) First 2-GHz Oscillator Phase Noise : Adding CL.

- Frequency was reduced from 22.6GHz to 1.93Ghz
- Phase noise should fall by $10\log(22.6\text{GHz}/1.93\text{Ghz})^2 = \mathbf{21.5\text{ dB}}$
- Power and Phase noise scale together
- FOM doesn't change

b) Second 2-GHz Oscillator Phase Noise : Increase No of stages.

- Frequency was reduced from 22.6GHz to 2.3Ghz.
- Flicker noise has improved. : $10\log(31/3) = \mathbf{10\text{dB}}$.
- Comparing with (a) : $10\log(2.3\text{ GHz}/1.93\text{ GHz})^2 = 0.8\text{ dB}$
- Expecting a **9.2B reduction** of Phase noise.
- FOM is improved as power consumption is independent of N.

*It's **better to increase the number of stages** than to **add capacitance**.*

c) Third 2-GHz Oscillator Phase Noise : Longer Length

- Weaker Inverters => low power due to less node capacitances.
- Obtained a frequency of ~2.5GHz.
- Phase noise is slightly higher than the increasing no. Of stages.
- **FOM is the highest : by 3-dB**
- Best phase noise-to-power ratio.

d) Fourth 2-GHz Oscillator Phase Noise : Use of frequency divider.

- O/p frequency is divided by M
- Phase is divided by M
- Phase noise is lower by a factor of M : **$20\log_{10}(M)$ in dB**
- No change in FOM
- No change in jitter as the divider o/p is also displaced by the same amt.

$$V_{div}(t) = V_0 \cos \left[\frac{\omega_0 t + \phi_n(t)}{M} \right]$$

Best solution is to increase the channel lengths of the transistors

Next best is to increase the No of inverters

Frequency tuning

Develop methods for tuning frequencies for the inverter-based ring oscillator

Tuning Considerations:

1. Tuning frequency range:

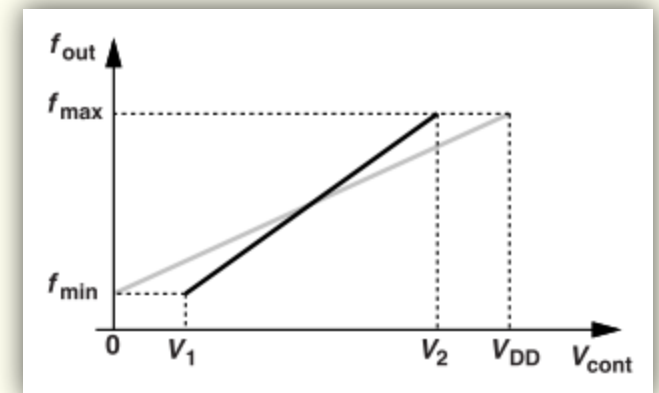
- Wide enough to cover both PVT variations & desired frequency range.

2. VCO gain KVCO.

- KVCO (Hz/v) makes VCO sensitive to noise on control line.
- A high value of Kvco is not desired as it indicates more freq change in shorter voltage control range. **KVCO must be low**
- A wide voltage control range allows us to keep lower KVCO.

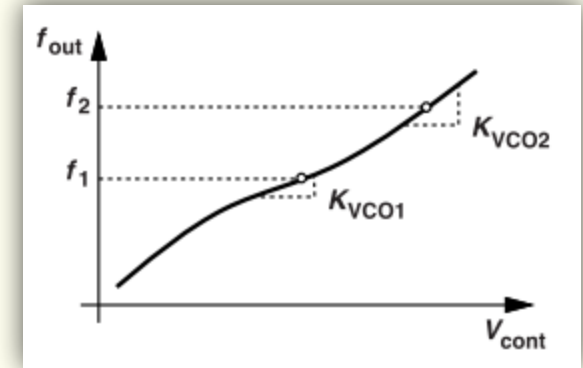
Rule of thumb: We keep KVCO below rough $0.1 f_m \text{ Hz/V}$

where ; $f_m = (f_{\max} + f_{\min})/2$



3. Linearity of tuning

- How much is the slope of KVCO varies?
- KVCO should vary by no more than 20% across the range.
- f_1 changes to f_2 : Expect KVCO2 to change not more than 20% of KVCO1



4. Constant Internal voltage swing.

- Prefer to keep the osc internal voltage swings constant across tuning range ; To avoid degrading the phase noise.
- Amplitude changes leak into phase noise. AM to PM conversion is min.

5. Stable Phase noise across range.

- Phase noise shouldn't change significantly as freq is tuned.

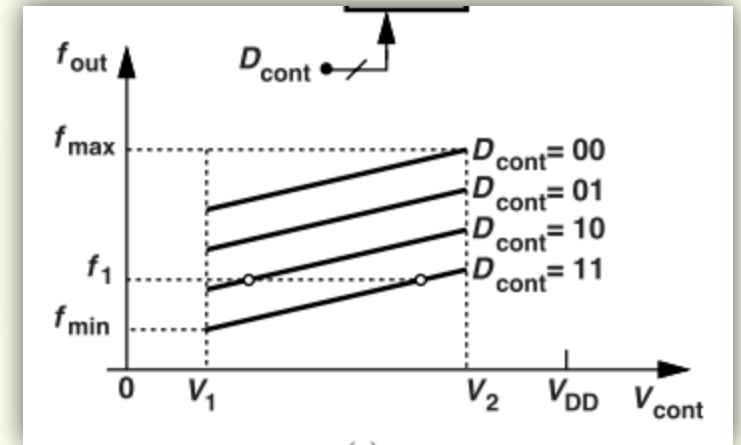
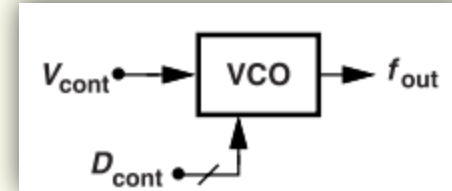
Continuous and Discrete tuning

- Larger frequency range make VCOs sensitive to noise on its control line.
- **Two types:** Analog-continuous tuning and Digital –discrete tuning.

IDEA :

- Choose a path controlled by **Weak Vcont**; to **lower the Kvco** and use **digital control** to jump across frequency bands to precisely control the tuning.
 - This is to get both **low phase noise** and **wide tuning range**.
1. **Continuous tuning** : Varies freq using Vcont (variable resistance)
 - Limited range to keep Kvco low.
 2. **Discrete tuning** : selects b/w the freq band.
 - By switching capacitors on/off.
 3. **Together** : Analog covers a small freq range for each digital setting, and the digital shifts the entire analog tuning curve into bits (1s and 0s).

- V_{cont} varies : $[v1, v2]$: small voltage range
- Entire Frequency range is utilized.

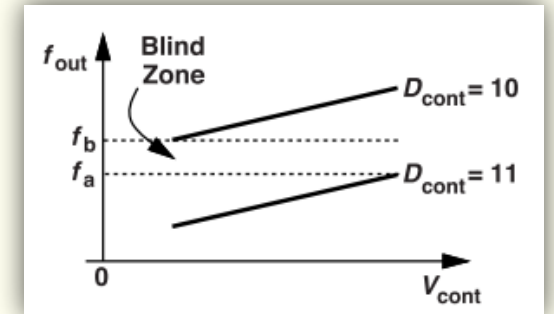


Blind Zone

- If two adjacent curves do not overlap,
- VCO fails to cover certain frequencies.

Rule of Thumb:

- Set VCO to cover 10% to 20% of center frequency.
- Choose no of steps for digital tuning to cover the full range.
- Increase the steps if KVCO is high.



Tuning by Variable Resistance

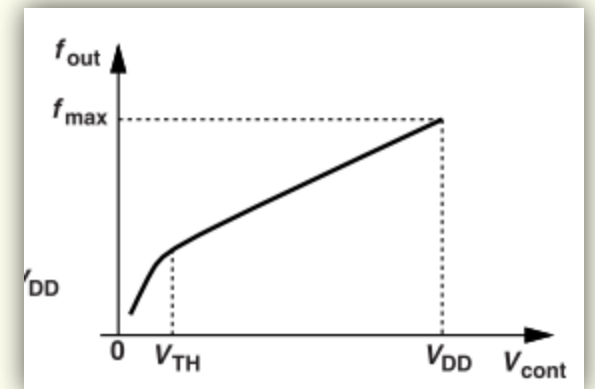
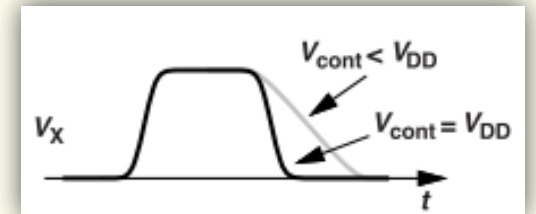
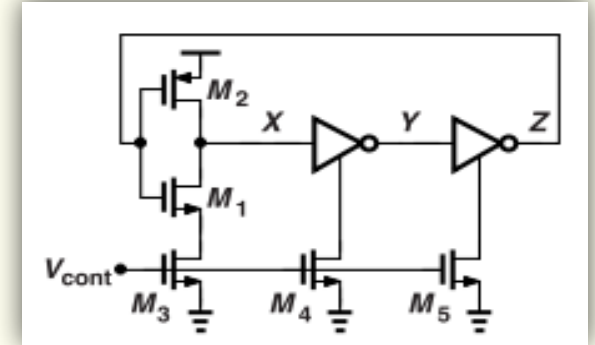
NMOS-based Variable resistance :

- M3-M5 : Voltage dependent resistors
- As $V_{cont} : [V_{DD} \rightarrow 0]$
 - At $V_{cont}=V_{DD}$; NMOS are shorted giving zero resistance and high freq.
 - As V_{cont} decreases, pull-down strength of inverters decrease.
 - At V_{cont} near to zero, M3 to M5 enter deep triode region.
 - Resistance go very high ; On-resistance increases.
 - Falling edge is slowed down.

Gate delay Increases leading to reduced Oscillation freq.

Problems??

- a) Asymmetric Transitions
- b) Non-linear tuning



Problems?

a) Asymmetric Transitions :

- Falling edge is slowed down due to increased resistance.
- Rising edge remains the same.
- This creates higher Phase noise on one side. (on the falling edge)
- At lower frequencies, it increases phase noise further.

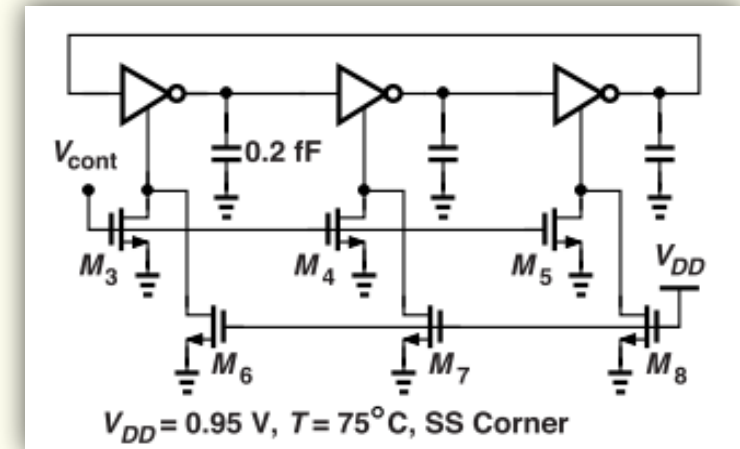
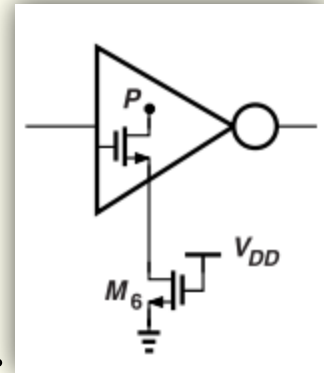
b) Non-linear tuning:

- When V_{cont} is near to zero, resistance increases sharply.
- This causes "sharp" drop in frequency. (as gate delay increases)
- *At $V_{cont}=0$; no oscillations .*

Solutions

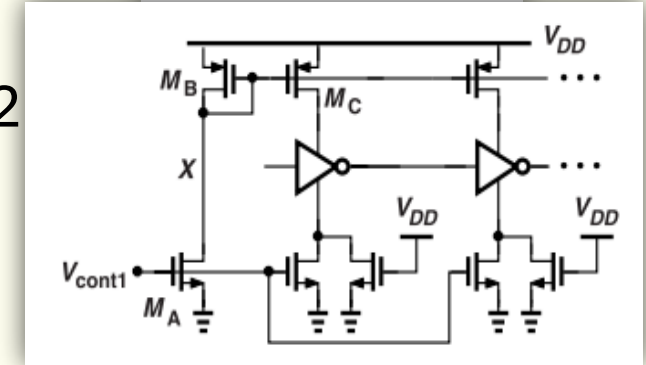
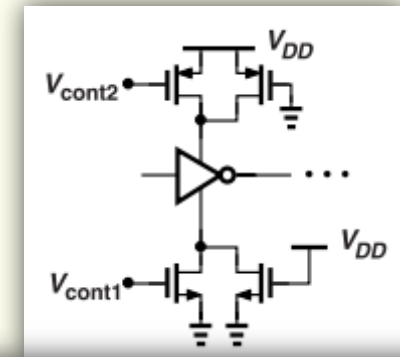
1. Solving Non-linear tuning:

- M6 ,M7, M8 transistors are added in parallel to M3, M4, M5.
- Ensures a min pull-down path at $V_{cont}=0$; Thus, oscillations sustain.
 - M6 to M8 never switch off ; connected to V_{DD} .
 - Provides a path to ground. O/p can still fall, and Oscillations sustain.
 - M6 to M8 are weak ; provide fixed resistance.
 - M3 to M5 are chosen strong maintain the maximum frequency.
- When $V_{cont} = V_{th}$;
 - M6 to M8 are in triode region.
 - They "degenerate" NMOS in the inverter.
 - This is increasing the NMOS (inverter) length.
 - They act in series thus **reducing Phase noise**.



2. Symmetric tuning

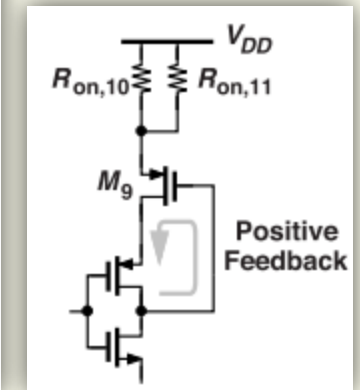
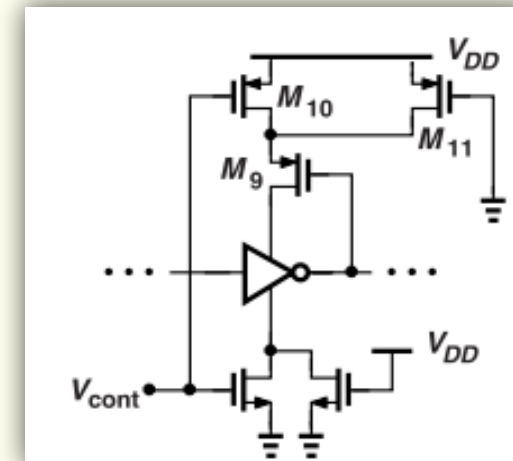
- Tune NMOS & PMOS strength for symmetric rise and fall.
- Increase V_{cont1} to weaken NMOS.
- Decrease V_{cont2} to weaken PMOS.
- Using current mirror: get V_{cont1} --> invert it --> get V_{cont2}
- The inverting stage (M_A and M_B) add Phase noise.
- Noise at node X corrupts the osc frequency.



Positive Feedback Tuning to reduce phase noise.

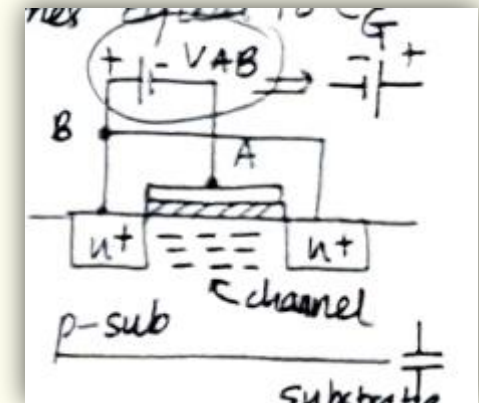
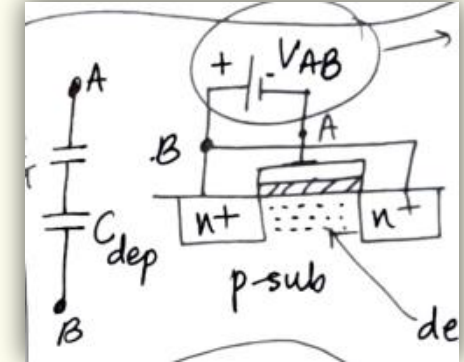
- M_9 PMOS as a feedback path. Controlled by V_{cont} through M_{10} .
- V_{cont} drops ; M_{10} PMOS turns more "ON".
- M_9 feedback strengthens and slows oscillator.

Achieves Symmetric Tuning



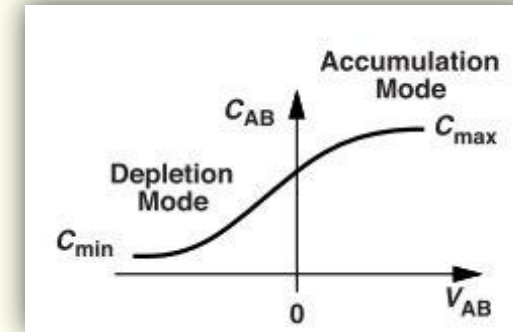
Tuning by Variable Capacitance

- Continuous tuning by means of voltage-dependent capacitors, also called “**varactors**”.
- If V_{ab} is negative:
 - Electrons are repelled and form a depletion region.
 - $C_{ab} = (C_g * C_{dep}) / (C_g + C_{dep})$;
- If $|V_{ab}|$ decreases:
 - Depletion region gets reduced.
 - Capacitance increases = higher C_{ab} .
- If V_{ab} is sufficiently positive:
 - Channel of electrons forms
 - $C_{ab} = C_g$



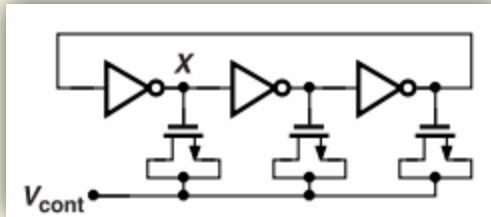
- **Modelling Varactor:**

$$C_{AB}(V_{AB}) = \frac{C_{max} - C_{min}}{2} \tanh \left(\alpha + \frac{V_{AB}}{V_0} \right) + \frac{C_{max} + C_{min}}{2},$$

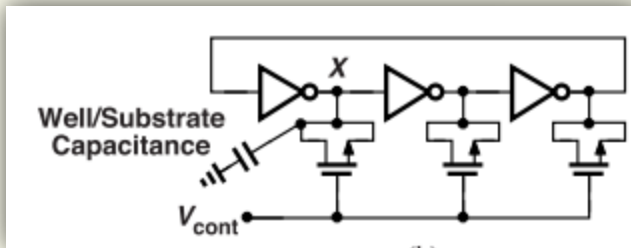


- **Integrating Varactors in Ring Oscillators**

a) $K_{vco} > 0$:



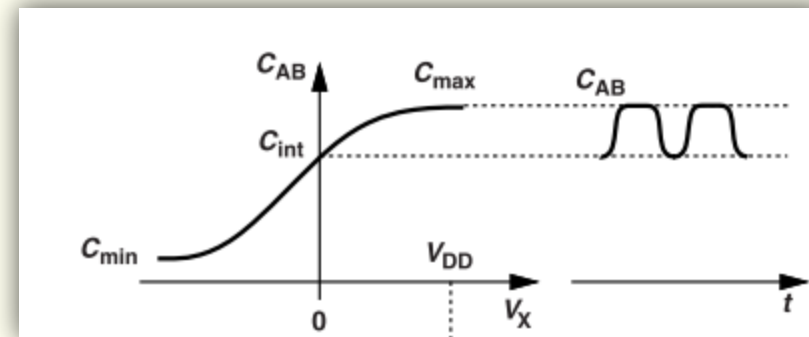
b) $K_{vco} < 0$:



- Tuning range is set by the varactor size
- **Advantages:**
 - **Very low noise compared to transistor tuning**

- **Behavior:**

- V_x swings from 0 to V_{DD}
- C_{ab} goes from C_{int} to C_{max} .
- C_{min} to C_{int} : not utilised as V_x can't be negative.



Discrete tuning

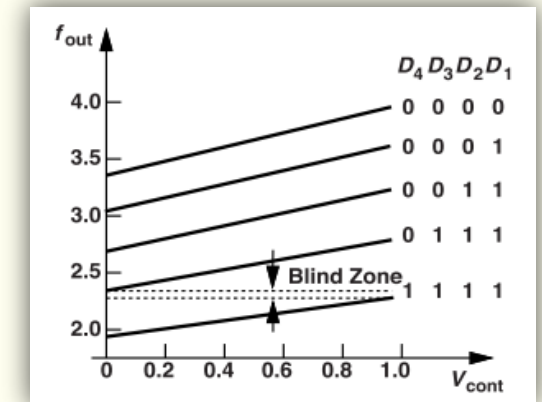
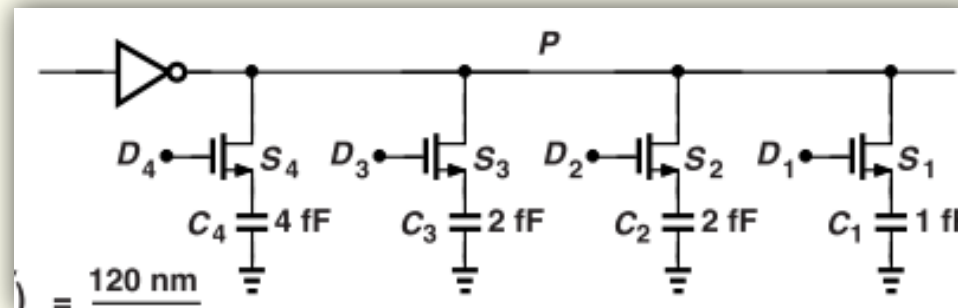
How is this implemented?

- Add programmable capacitors at each node. { C1 to C4 }
- The switches are driven by "thermometer" code.
- As more switches are "on" : C increases and f decreases.

How to select Capacitance values?

- C1: smallest ; C4: largest and C3=C2

$D_4 D_3 D_2 D_1$	Load Capacitance
0 0 0 0	0
0 0 0 1	1 fF
0 0 1 1	3 fF
0 1 1 1	5 fF
1 1 1 1	9 fF



Problem of supply Noise

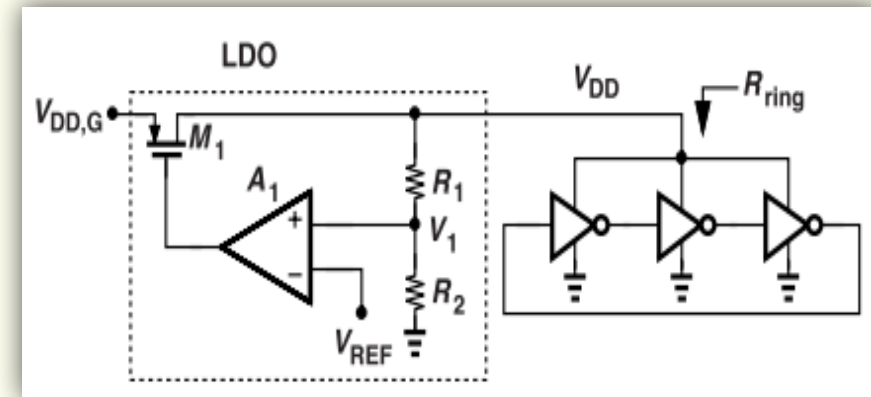
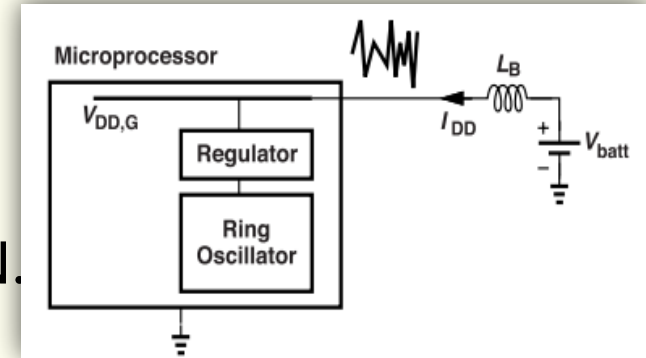
- Ring osc. are sensitive to supply voltage
- VDD noise modulates directly with frequency of osc and PN.

Use of a Regulator : to mitigate noise

- Regulators o/p voltage is lower than the global VDD.
- Parasitic Inductor L_B current change translates to large voltage change thus disturbing oscillator.

A. Voltage Regulation- LDO

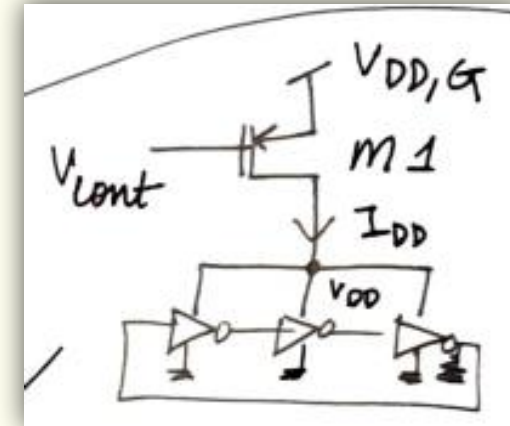
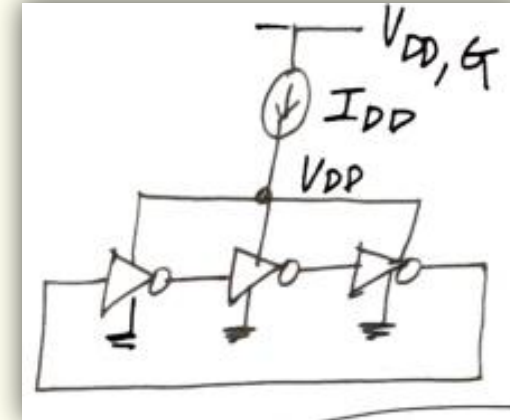
- LDO generates a stable VDD for ring Osc.
- A1 ensures $V_1 = V_{ref}$ and **$VDD = V_{ref} (1 + R_1/R_2)$ is regulated.**



LDO has good supply noise rejection and tunable output range

B . Current Regulation

- Isolate the ring osc from supply.
- Add a Current source in between.
- I_{dd} consumes some voltage headroom.
- $V_{dd} = I_{dd} / (3 \cdot f_0 \cdot C_{tot})$
- *Change f_0 using current*
- Noise in $V_{dd,g}$ modules V_{gs} and I_{DD}
- M1 has flicker and thermal noise.
- V_{cont} can't reach $V_{dd,g} \rightarrow m1$ off.



Using a current mirror gives worse tuning range and noise than LDO.